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RESEARCH ARTICLE

Lightweight Forensic Triage Framework for Clandestine Operational Instructions: A Two-Case Leave-One-Case-Out Evaluation

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ABSTRACT Digital forensic casework rarely proceeds under laboratory conditions. Under time and resource constraints, analysts often triage thousands of seized text files, with critical documents embedded in large volumes of irrelevant material. This study presents a lightweight triage approach for prioritizing clandestine operational instruction (COI) documents in CPU-only deployment environments. The method combines TF-IDF features, a LinearSVC classifier, a tunable decision threshold, and SHAP-based analyst-facing rationale. Cross-case transfer was evaluated with a leave-one-case-out (LOCO) protocol using 133 court-adjudicated COI documents from two independent investigations and 500 synthetic negatives constructed as a structured stress-test contrast set. Across the two folds, the model achieved a mean accuracy of 99.29% and a mean F1 of 98.28%, with 5 false negatives in total (fold-wise FNR: 2.27% and 4.49%; pooled FNR: 3.76%). End-to-end screening required 0.32 ms per document, with a 1.34 MB model footprint on a single-threaded CPU. Comparisons with transformer-based and lightweight baselines showed that this configuration provided a favorable balance of accuracy, latency, model size, and feature-level auditability under CPU-constrained conditions. A proxy false-positive evaluation on 9,606 Korean Wikipedia paragraphs from KorQuAD v1.0 yielded a false-positive rate of 0.31% (95% CI: 0.21%–0.45%). Work Saved over Sampling reached 79%. The system is intended as a screening aid within a tiered forensic workflow. SHAP outputs are designed to support analyst review rather than serve as standalone evidentiary instruments. The findings are limited to the evaluated Korean-language counterintelligence cases and have not been validated across other languages, crime types, or jurisdictions.

INDEX TERMS Artificial intelligence, decision support systems, digital forensics, explainable artificial intelligence, natural language processing, support vector machines.

I. INTRODUCTION

A. OPERATIONAL MOTIVATION

Digital forensic investigations typically do not unfold in small, orderly collections [1]. In practice, analysts are confronted with thousands of text files—emails, notes, draft reports, and fragments of correspondence—extracted from seized storage devices, most of which have no evidentiary

value [2]. The manual review of the entire corpus is slow and, in some cases, strategically hazardous [3]. The most consequential documents are often rare. They can be obscured within a large volume of irrelevant material [4].

For this reason, automated text screening—commonly termed forensic triage—has become a practical first stage in digital evidence workflows. The objective is not to replace human judgment but to reorder the review process. A lightweight classifier assigns a relevance score to each

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document, enabling analysts to prioritize materials that warrant closer examination [5], [6].

This study applies this triage principle to a narrowly defined yet operationally significant document category: clandestine operational instructions (COI). COI documents are directive texts used to coordinate covert activities and are typically written in a standardized, template-like style. Although the corpus is derived from publicly adjudicated counterintelligence cases in South Korea [7], [8], the underlying technical problem is not jurisdiction-specific. This study identifies low-frequency, formulaic directive documents embedded within large seized text corpora.

In such settings, scarcity rather than scale defines the challenge. The critical text is rare and dispersed within substantial background material. The principal constraint is not computational capacity but analyst time.

COI documents frequently contain operational directives aligned with political or organizational objectives [9]. A prior linguistic analysis of 133 authentic COI texts from two independent cases [9] documented recurring textual regularities. These included the use of coded vocabulary—such as corporate metaphors to describe hierarchical relations—as well as a recurrent rhetorical structure consisting of an opening salutation, situational framing, operational directives, and a formal closure. Thematic emphasis was consistently placed on political mobilization and electoral coordination. These regularities indicate the persistence of a shared lexical and structural pattern across cases and time periods. This recurring configuration of expressions and cues forms a consistent lexical core.

B. PROBLEM STATEMENT AND DESIGN GOALS

Counter-intelligence investigations generate substantial volumes of textual evidence, frequently amounting to thousands of files per case [10]. Exhaustive manual review is time-intensive and increases the risk of undetected low-frequency yet operationally consequential documents [11]. Tools specifically tailored to clandestine operational instructions remain uncommon within routine forensic workflows [12].

Therefore, the technical objective is defined in operational terms: to determine whether a compact model can prioritize COI documents across independent investigations without exceeding practical deployment constraints. The proposed system is designed with four explicit requirements. It must operate on central processing unit (CPU) without GPU acceleration. The inference latency must remain at the millisecond scale per document. Explanations must be available at the feature level to support analyst verification. The cross-case transfer must be evaluated under strict case-level separation.

The consistent lexical core hypothesis assumes that recurring structural and lexical patterns persist across investigations and can be represented through sparse feature encoding. This assumption is examined by comparing a term frequency-inverse document frequency representation

combined with a linear support vector machine against alternative baseline models. Performance is evaluated using a Leave-One-Case-Out (LOCO) protocol, where training is conducted on one investigation and testing is conducted on the other. This design isolates genuine cross-case transfer from effects of within-case similarity.

C. CONTRIBUTIONS

The contribution of this work lies in system design, validation methodology, and deployment characterization rather than algorithmic novelty alone:

(1) Framework integration: A four-stage, CPU-only, end-to-end triage pipeline combining sparse TF-IDF feature extraction, threshold-configurable LinearSVC classification, SHAP-based analyst rationale, and chain-of-custody reporting is implemented and evaluated as an operational unit.

(2) Cross-case validation protocol: Transfer performance is assessed under strict case-level leave-one-case-out (LOCO) separation, preventing cross-contamination that occurs when documents from the same investigation appear in both training and test splits. Deployment behavior is further examined through a structured synthetic contrast set, a proxy false-positive evaluation on 9,606 KorQuAD [13] paragraphs, and triage-centric metrics including Recall@K and Work Saved over Sampling ($WSS@100\% = 79\%$) [14].

(3) Error profile reporting: False-negative rates are reported per fold (2.27%, 4.49%) alongside qualitative analysis of missed positive examples, addressing the forensic primacy of recall over precision in triage settings.

(4) Lightweight baseline comparison: Performance is benchmarked against fastText [15] and KoELECTRA-small [16], [17] under identical LOCO evaluation, demonstrating that the TF-IDF+LinearSVC configuration achieves the optimal deployment trade-off among accuracy, latency (0.32 ms), model size (1.34 MB), and sparse-feature auditability.

(5) Preprocessing evaluation: Whitespace tokenization is evaluated against Korean morphological analysis (Okt), showing that morphological analysis improves accuracy to $F1 = 1.0$ but incurs 143x latency overhead, validating the whitespace design for latency-critical deployment.

(6) Explainability governance: SHAP is framed and documented as an analyst-facing decision-support tool with explicit limitations regarding evidentiary use.

II. BACKGROUND AND RELATED METHODS

A. DIGITAL FORENSIC TRIAGE

Digital forensic triage has emerged as a response to the increasing scale of digital evidence. Early approaches emphasized tiered screening, in which lightweight automated procedures precede comprehensive manual examination [5], [6], [18]. Kent et al. [5] described triage as an ordering strategy for evidence review tasks. Jusas et al. [6] surveyed digital triage tools and underscored the role of rapid first-pass filtering. Hitchcock et al. [18] described a field-oriented

methodology to support nonspecialist personnel during initial evidence handling.

Subsequent research has extended triage concepts into more formal settings. Sandvik et al. [19] examined the related principles within fog computing infrastructures. In IEEE Access, Alotibi [20] proposed a digital forensic readiness metamodel addressing organizational preparedness, while Oh et al. [21] investigated timestamp manipulation detection within forensic workflows.

Text-specific triage under strict case-level separation remains comparatively underexplored. Existing studies often focus on artifact categorization or infrastructure readiness rather than prioritizing cross-case documents. Fewer studies have assessed whether prioritization models retain performance across independent investigations while satisfying deployment constraints. This condition is evaluated directly through resource-constrained text triage with explicit cross-case validation.

B. STYLOMETRIC REGULARITY AND DIRECTIVE TEXT

Stylometry is based on the premise that written texts have measurable and recurrent stylistic features. Authorship attribution infers authorship from stylometric and lexical signals and is well established in digital forensics. Stamatatos [22] synthesized various statistical and machine learning techniques for attribution tasks. Recent evaluation frameworks indicate that lexical frequency-based features and linear classifiers can retain competitive discriminative performance across author sets of varying size [23]. Author profiling extends this line of work by inferring demographic or behavioral characteristics from textual traces [24].

Stylometric reasoning has also been applied beyond individual attribution. Studies of institutional discourse, particularly within authoritarian regimes, have documented official texts' constrained linguistic variation and recurring formal structures. Kim [25] observed recurring rhetorical templates in North Korean New Year addresses. Baek [26] examined the compositional conventions in Rodong Sinmun. Jung [27] presented longitudinal evidence that institutional texts in North Korea preserve formulaic patterns over extended time spans. Ko and Kim [9] provided a more direct precedent for this investigation. By analyzing 133 court-verified COI documents with TF-IDF features, n-grams, and BERT-based clustering, they found marked lexical concentration and structural regularity within the directive texts. However, that analysis remained primarily descriptive. The cross-case transfer and operational triage performance were not evaluated. This study reconsiders the same empirical setting under a different criterion: whether recurrent institutional regularity can support reliable document prioritization across independent investigations.

C. EXPLAINABLE MACHINE LEARNING FOR FORENSIC SUPPORT

Interpretability carries particular weight in forensic workflows [28] [29]. SHAP [30] offers a principled mechanism

for decomposing a prediction into feature-level contributions by estimating marginal effects. In linear models, these quantities directly align with weighted term coefficients, enabling the transparent identification of lexical drivers behind each classification outcome [31].

The need for complex neural architectures in specialized forensic domains has also been questioned. Wahba et al. [32] observed that traditional machine learning approaches, when coupled with domain-specific feature engineering, can match or exceed transformer-based models in narrowly defined classification tasks. Studiawan et al. [33] emphasized that evaluation rigor, rather than architectural novelty, determines the reliability of digital forensic research. Göbel et al. [34] examined the controlled use of synthetic data and argued that under security and access constraints, carefully constructed synthetic corpora can complement authentic evidence.

Broader natural language processing research has explored instruction-aligned large language models and neural topic modeling frameworks for representation and corpus construction. Self-Instruct [35] illustrates how instruction-driven data generation can expand task-specific datasets. BERTopic [36] combined transformer embeddings with class-based TF-IDF to reveal latent semantic structure. High-dimensional lexical modeling remains competitive in large-scale authorship attribution and text classification contexts due to consistency and interpretability [37]. This framework is situated within this landscape but adopts a narrower emphasis. The primary design criteria are computational parsimony, explicit feature attribution, and cross-case validation. The architectural complexity is not excluded; it is simply not assumed to be necessary under the evaluated constraints.

To summarize, existing forensic triage literature addresses artifact categorization and infrastructure readiness but rarely evaluates text-level prioritization under strict case-level separation. Stylometric studies document lexical regularity in institutional texts yet do not operationalize these findings for automated screening. Explainable ML frameworks for forensic contexts have been proposed at the architectural level, but no prior work integrates CPU-deployable sparse classification, case-level cross-validation, threshold-tunable triage, and feature-level explainability within a single evaluated pipeline. The present study addresses this gap. Prior forensic NLP studies often rely on relatively large annotated corpora or are evaluated on English-language datasets. This work examines low-resource Korean document triage using a minimal feature set with LOCO validation across independent investigations.

III. PROPOSED LIGHTWEIGHT TRIAGE FRAMEWORK

A. SYSTEM REQUIREMENTS

Digital forensic triage systems are typically deployed in environments where computational resources are limited and rapid screening is required [38]. Therefore, the proposed framework is designed around explicit operational constraints rather than solely maximizing predictive accuracy. In this

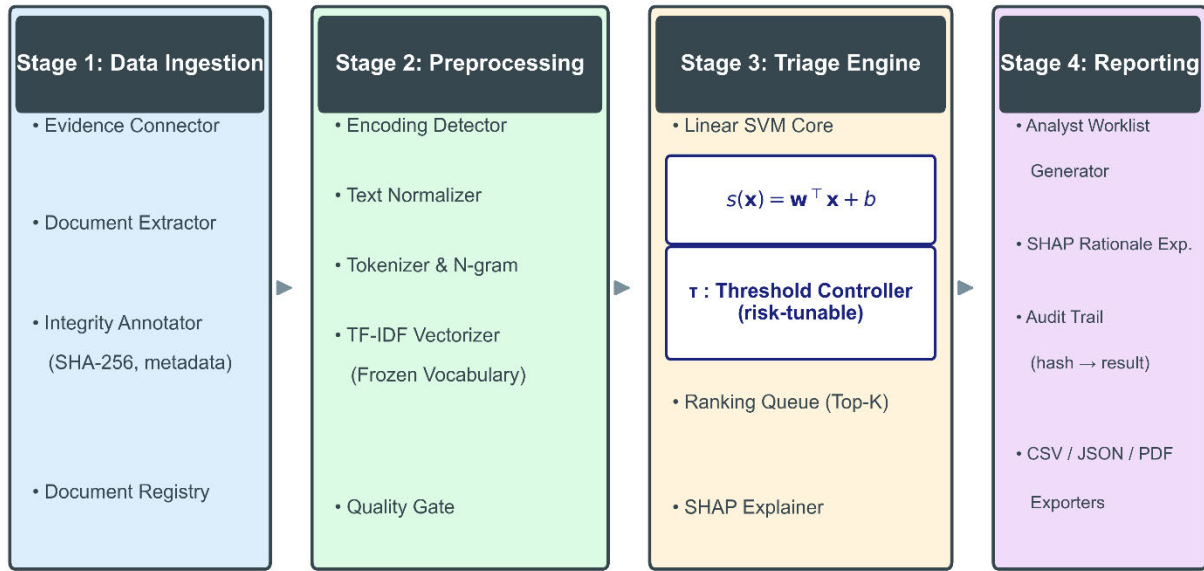


FIGURE 1. The system architecture of the proposed forensic triage pipeline. Four stages process seized documents from ingestion to reporting.

context, the objective is to balance prioritization reliability with predictable execution behavior under fixed hardware conditions.

First, the system must operate entirely on a CPU without relying on graphics processing hardware. Several investigative workstations exclude dedicated accelerators, and deployment should not depend on specialized infrastructure [39]. Second, preprocessing must be deterministic. The vocabulary is frozen after training so that feature mappings remain consistent across repeated executions and investigative cases. Third, all experiments and runtime measurements were conducted in single-threaded mode to ensure reproducibility and consistent latency estimation. Fourth, inference time must remain at the millisecond level per document so that thousands of files can be screened within practical time constraints. Finally, the system must provide feature-level explanations, enabling analysts to identify the specific terms that contributed to a document being flagged [40].

The intended deployment environment is a standard forensic workstation that processes large volumes of text extracted from seized digital storage devices. In such settings, screening precedes in-depth analysis and often determines the investigative direction. Under these conditions, computational efficiency, determinism, and interpretability are treated as primary design parameters rather than secondary optimizations.

B. ARCHITECTURE OVERVIEW

Fig. 1 presents the system architecture, structured as four sequential stages: data ingestion, preprocessing, triage engine, and reporting. The stages are executed in order,

but the design principle is consistency rather than complexity. Each component preserves determinism and traceability across the workflow.

Stage 1 (Data Ingestion) registers documents extracted from seized storage media, computes cryptographic hashes, and associates metadata required for chain-of-custody (COC) documentation [41]. This step establishes an integrity baseline before any analytical transformation occurs.

Stage 2 (Preprocessing) converts the raw textual content into numerical feature representations. Encoding detection and normalization are applied before tokenization and n-gram generation. TF-IDF vectorization is performed using a frozen vocabulary to ensure that feature indices remain consistent across executions and investigative cases.

Stage 3 (Triage Engine) computes classification scores using a linear support vector machine defined as $s(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$. A configurable threshold τ governs operational sensitivity and allows risk-adjustable prioritization. Documents exceeding the threshold are ranked and queued for analyst review. At this stage, SHAP-based explanations are generated to expose the feature contributions underlying each decision [42].

Stage 4 (Reporting) produces analyst-facing outputs, such as prioritized worklists, feature rationale exports, and audit trails linking document hashes to prediction outcomes. Structured report formats (CSV, JSON, and PDF) are generated for documentation and downstream integration.

C. PREPROCESSING MODULE

The preprocessing module transforms the extracted text into a sparse TF-IDF representation. Let d denote a seized

document drawn from corpus D . A deterministic mapping g converts each document into a vector in a p -dimensional feature space,

$$g : D \rightarrow \mathbb{R}^p, x = g(d) \quad (1)$$

where p corresponds to the fixed vocabulary size.

The vectorizer employs unigram and bigram features (ngram range 1–2), a minimum document frequency of 2, a maximum document frequency of 0.95, and sublinear term-frequency scaling. Once training is complete, the vocabulary remains frozen to preserve feature index alignment across repeated executions and investigative cases.

No language-specific morphological analyzer is used. This is an intentional design decision. Within the target domain, the COI documents exhibit rigid and template-like surface forms. Whitespace-level tokenization, implemented via Python's default `str.split()`, retains discriminative lexical markers while avoiding additional latency and external software dependencies [43]. The resulting pipeline remains deterministic and lightweight under CPU-only deployment.

D. TRIAGE ENGINE MODULE

The triage engine implements the following binary decision rule:

$$f(x) = \mathbb{I}[s(x) \geq \tau] \quad (2)$$

where the decision score is defined as follows:

$$s(x) = w^T x + b \quad (3)$$

A document is labeled COI-positive when $s(x) \geq \tau$, and COI-negative otherwise. The default setting $\tau = 0$ corresponds to the decision boundary of the canonical support vector machine. In operational use, τ may be adjusted to shift the recall–precision balance according to investigative priorities.

The weight vector w is obtained by solving the following soft-margin optimization problem:

$$\begin{aligned} \min \frac{1}{2} \|w\|^2 + C \sum_i \alpha_{y_i} \xi_i \\ \text{s.t. } y_i(w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0 \end{aligned} \quad (4)$$

where α_{y_i} denotes class-specific weights (implemented using balanced class weighting), $C = 1.0$ is the regularization parameter, and ξ_i are slack variables.

The resulting linear decision surface is compatible with high-dimensional sparse text features and remains computationally efficient during inference.

E. EXPLAINABILITY MODULE

The system produces two complementary forms of explanation for each document classified as COI-positive.

The first is a lightweight, score-based explanation. For a linear model, each feature's contribution is directly proportional to the product $w_j x_j$. The top- k contributing terms are therefore identified by ranking these weighted feature values,

allowing analysts to quickly determine which tokens most strongly influenced the decision.

The second level employs SHAP [30], which express the prediction as an additive expansion, as follows:

$$F(x) = \phi_0 + \sum_{j=1}^p \phi_j(x) \quad (5)$$

where $\phi_j(x)$ denotes the contribution of feature j for document x , and ϕ_0 represents the expected model output.

To estimate the corpus-level feature influence, we compute

$$I_j = \left(\frac{1}{N} \right) \sum_{i=1}^N |\phi_j(x_i)| \quad (6)$$

where N is the number of documents. This quantity reflects how consistently feature j contributes to model decisions across the evaluated corpus.

F. REPORTING AND GOVERNANCE MODELS

The reporting module converts classification outputs into artifacts that are analyst-facing. Documents are ranked by decision score and assembled into a prioritized worklist. For each flagged item, the SHAP-based summaries expose the primary lexical contributors to the prediction. The system logs document hashes with classification outcomes to maintain traceability during review. Structured exports (e.g., CSV and JSON) enable interoperability with external forensic tools.

Governance controls are implemented to maintain consistency over time. The vocabulary version is fixed and recorded. The LOCO validation reports are archived. Runtime profiling logs document the latency and execution conditions. Analyst-confirmed labels may be incorporated during scheduled retraining procedures, subject to controlled evaluation. Validation processes are executed separately from runtime inference to prevent unintended drift during operational deployment.

IV. DATASET AND STRESS-TEST DESIGN

A. AUTHENTIC COI CORPUS

The corpus comprises 133 authentic COI documents extracted from court-adjudicated national security cases. Materials were obtained from the Korean Supreme Court Legal Information System and the legal database [7], [8]. To ensure compliance with ethical and legal constraints, all personally identifiable information was removed prior to analysis [44]. IRB approval was not required because this study used only publicly available materials, including judicial records from the Korean Supreme Court Legal Information System and the KorQuAD v1.0 corpus. No individuals were contacted or recruited.

These two investigations correspond to publicly adjudicated COI cases with full-text directive documents accessible through the Korean legal information system at the time of this study. Other counterintelligence cases may be sealed, involve classified materials, or lack digitized evidentiary records, constraining broader data access.

TABLE 1. Overview of the dataset.

Category	Documents	Period	Source	Mean Length (chars)	Std
Case A	44	2018–2021	Court records [7]	2,262	1,077
Case B	89	2018–2022	Court records [8]	2,548	1,893
Authentic COI (pooled)	133	2018–2022	—	2,422	1,648
Synthetic Negative	500	N/A	Claude Sonnet 3.7	1,098	496

Table 1 summarizes the structure of the dataset. Case A contains 44 documents associated with an investigation conducted between 2018 and 2021 [7]. Case B includes 89 documents from a separate investigation spanning 2018–2022 [8]. The two cases involved distinct operational groups and were investigated in different geographic regions. This separation provides a basis for evaluating transferability beyond the lexical similarity of within-case cases.

To construct a contrast class under controlled conditions, we generated a synthetic negative set of 500 texts using Claude Sonnet 3.7. These texts were formatted to resemble administrative or organizational communication but did not have the directive content characteristic of COI documents. The synthetic set functions as a structured stress-test complement rather than a substitute for authentic evidence. Therefore, the complete dataset comprises 633 documents.

B. SYNTHETIC NEGATIVE CLASS

The negative class consists of 500 LLM-generated texts formatted as organizational correspondence. These texts were generated to avoid directive-specific vocabulary while preserving surface characteristics of organizational documents. The synthetic set serves as a controlled contrast class for stress-testing classifier boundaries rather than as a representative sample of real seized material.

Generation prompts constrain format and length but are designed to avoid terms associated with operational coordination. As shown in Table 1, the pooled mean document length of authentic COI texts (2,422 characters; $SD = 1,648$) exceeds that of synthetic negatives (1,098; $SD = 496$). The authentic corpus exhibits high variance, particularly in Case A (mean 2,262; $SD = 1,077$), with the lower tail extending below 1,200 characters, resulting in overlap with the synthetic length distribution. The median COI length (1,929 characters) is lower than the pooled mean, reflecting the influence of longer documents in Case B. This asymmetry is consistent with the data construction: shorter documents tend to produce sparser TF-IDF vectors with fewer nonzero features, making the classification boundary more conservative rather than more permissive. Classification performance does not appear to be strongly driven by document length, as suggested by the proxy false-positive evaluation (Section V-D), in which 9,606

KorQuAD paragraphs, filtered only by a 200-character minimum, produced an FPR of 0.31%. This restriction reduces topical leakage and enables the evaluation of whether the model depends on COI-specific lexical structure rather than superficial formatting cues.

This approach is consistent with established digital forensic methodologies, in which synthetic corpora are used to simulate controlled negative conditions when authentic data access is restricted [34], [45]. This study’s central claim does not rely on synthetic performance. Cross-case lexical regularity is evaluated exclusively using authentic COI documents and strict separation across independent investigations.

C. LOCO VALIDATION PROTOCOL

Conventional k-fold cross-validation distributes documents from both investigations across training and test folds. Although this procedure estimates the aggregate predictive accuracy, it does not isolate the transfer across investigations. Documents originating from the same case may share lexical templates or boilerplate phrasing, and these patterns can appear in both splits. Under such conditions, the k-fold evaluation may inflate the apparent generalization [46].

The validation strategy adopted here follows a LOCO protocol intended to approximate deployment conditions [47]. The model is trained on COI documents from one adjudicated investigation and evaluated on an entirely separate investigation. This case-level partition prevents within-case leakage and tests whether a compact lexical representation transfers across independent operational groups.

Positive documents remain strictly case-separated. Negative samples are proportionally partitioned to preserve class balance without introducing cross-case contamination. In LOCO-1, training is conducted on Case B (89 positive documents) and testing is conducted on Case A (44 positive documents). LOCO-2 reverses this configuration.

Table 2 summarizes the experimental configuration. The enforced case-level separation ensures that reported performance reflects cross-case transfer rather than intra-case similarity.

TABLE 2. LOCO cross-validation configuration.

Fold	Train Positive	Train Negative	Test Positive	Test Negative
LOCO-1	Case B (89)	334	Case A (44)	166
LOCO-2	Case A (44)	165	Case B (89)	335

D. EXPERIMENTAL ENVIRONMENT AND REPRODUCIBILITY

All experiments were conducted on a workstation equipped with an AMD Ryzen 5-class processor and 32 GB of RAM, operating under Windows 11 (64-bit). Random seeds were fixed (`random_state = 42`) to ensure deterministic model training and evaluation. The TF-IDF vocabulary was frozen after training so that feature indices remained consistent across executions and validation folds. Model artifacts—including the vectorizer and classifier

parameters—were serialized to preserve exact configuration reproducibility.

Latency benchmarking was performed using single-threaded execution. The environment variables (e.g., `OMP_NUM_THREADS = 1` and `MKL_NUM_THREADS = 1`) were set to eliminate the variability introduced by parallel scheduling. A brief warm-up phase preceded the measurement, and the inference time was averaged over 100 iterations using `time.perf_counter`.

Implementation was carried out in Python 3.9.21 using widely adopted machine-learning libraries: scikit-learn 1.5.2 (LinearSVC, LogisticRegression, MultinomialNB, MLPClassifier, TfidfVectorizer), XGBoost 2.1.3, SHAP 0.47.2, NumPy 1.26.4, pandas 2.2.3, and SciPy 1.13.1. Additional lightweight baseline experiments used fastText for bag-of-n-gram classification, and Korean morphological tokenization was implemented with the Okt analyzer. Deep-learning baselines employed the pre-trained jinaai/jina-embeddings-v3 model (XLM-RoBERTa-based) and the monologg/koelectra-small-v3-discriminator checkpoint via Hugging Face Transformers 4.52.3 and PyTorch 2.6.0 (CUDA 11.8).

Graphics acceleration was disabled for all efficiency measurements to align with the intended deployment environment (`CUDA_VISIBLE_DEVICES = ""`).

For the Jina v3 baseline, the embeddings used in the LOCO validation were pre-computed and cached. This step does not affect the classification outcomes because the embedding computation is deterministic once the model parameters are fixed. All latency measurements, including end-to-end and inference-only benchmarks, were conducted under CPU-only execution, consistent with standard forensic workstation conditions. The same LOCO protocol and single-threaded CPU setting were also applied to the fastText and KoELECTRA-small baselines and to the tokenizer ablation comparing whitespace tokenization with Okt morphological analysis.

V. EXPERIMENTAL EVALUATION

A. BASELINES AND CONFIGURATIONS

Five text classification configurations are examined to assess the proposed framework under consistent feature settings.

The primary model combines TF-IDF features [48] representation with a linear support vector machine [49] ($C = 1.0$, balanced class weights, `max_iter = 10,000`). Logistic regression [49] is evaluated using the same feature representation to isolate the effect of the decision function ($C = 1.0$, balanced class weighting, L-BFGS solver). A multinomial Naive Bayes classifier with smoothing parameter $\alpha = 1.0$ is included to represent a linear probabilistic baseline.

TF-IDF features are further paired with an XGBoost classifier to examine nonlinear decision boundaries under identical sparse inputs (`n_estimators = 100`, `max_depth = 6`). An embedding-based baseline is also considered. Jina v3 [50], a multilingual transformer encoder grounded in large-scale pretraining paradigms widely used in NLP [51],

generates 1024-dimensional document representations. These embeddings are passed to a multilayer perceptron with two hidden layers (256 and 128 units), ReLU activation, and early stopping.

To contextualize the proposed TF-IDF+LinearSVC configuration, Table 3 compares three lightweight baselines under identical LOCO evaluation. The fastText (773 MB, $F1 = 0.9769$) offers competitive accuracy with faster inference (0.16 ms) but provides limited sparse-feature traceability; it was trained with 25 epochs, learning rate 0.5, word n-grams of order 2, embedding dimension 100, and softmax loss. KoELECTRA-small — employing the monologg/koelectra-small-v3-discriminator pretrained checkpoint with a logistic regression head ($C = 1.0$, balanced class weighting) — (55 MB, $F1 = 1.0$) achieves perfect recall but incurs 9.3 ms latency, $29\times$ slower than the proposed architecture. The TF-IDF+LinearSVC configuration is selected as the deployment core based on the practical balance among accuracy, latency (0.32 ms), model size (1.34 MB), and sparse-feature auditability. Although KoELECTRA-small achieved $F1 = 1.0$ across the two LOCO folds, this result reflects zero false negatives among 133 positive test instances and does not statistically guarantee $FNR = 0$ under broader deployment. Moreover, the model provides less direct sparse-feature-level traceability, which may limit per-token rationale in analyst-facing triage documentation.

TABLE 3. Comparison of lightweight baselines under LOCO evaluation.

Model	Mean F1	Total FN	Latency (ms)	Size (MB)
TF-IDF+LinearSVC	0.9828	5	0.32	1.34
fastText	0.9769	6	0.16	773
KoELECTRA-small	1.0	0	9.3	55

B. CROSS-CASE PERFORMANCE UNDER LOCO VALIDATION

The standard k-fold validation distributes documents from both investigations across training and test splits. When case-specific templates appear in both subsets, the apparent generalization may be inflated. Therefore, all models are evaluated under the LOCO protocol to isolate cross-case transfer.

Table 4 presents the LOCO results for both folds. Performance varies by the model and fold.

Under LOCO-1, linear models (SVM and logistic regression) produce near-identical outcomes with minimal and no false negatives. Multinomial Naive Bayes shows reduced recall. XGBoost remains competitive in LOCO-1 but exhibits degradation under LOCO-2 conditions.

In LOCO-2, TF-IDF + LinearSVC and TF-IDF + logistic regression again maintained high recall with zero false positives. In contrast, XGBoost yields a substantial drop in recall (0.3034) and increases false negatives (62). The embedding-based Jina v3 + MLP baseline improves recall relative to Naive Bayes but introduces additional false

TABLE 4. LOCO cross-validation results.

Model	Fold	Acc.	Prec.	Rec.	F1	FP	FN
TF-IDF + LinearSVC	LOCO-1	0.9952	1.0000	0.9773	0.9885	0	1
TF-IDF + LogReg	LOCO-1	0.9905	1.0000	0.9545	0.9767	0	2
TF-IDF + MultinomialNB	LOCO-1	0.9571	1.0000	0.7955	0.8861	0	9
TF-IDF + XGBoost	LOCO-1	0.9762	1.0000	0.8864	0.9398	0	5
Jina v3 + MLP	LOCO-1	0.9905	1.0000	0.9545	0.9767	0	2
TF-IDF + LinearSVC	LOCO-2	0.9906	1.0000	0.9551	0.9770	0	4
TF-IDF + LogReg	LOCO-2	0.9906	1.0000	0.9551	0.9770	0	4
TF-IDF + MultinomialNB	LOCO-2	0.9788	1.0000	0.8989	0.9467	0	9
TF-IDF + XGBoost	LOCO-2	0.8514	0.9643	0.3034	0.4615	1	62
Jina v3 + MLP	LOCO-2	0.9528	0.9481	0.8202	0.8795	4	16

positives and higher computational cost. The TF-IDF + LinearSVC configuration maintains the most consistent balance between recall and false-positive control across folds. Errors are confined to some missed positives, whereas spurious activations in the negative class are not observed under LOCO conditions.

Logistic regression produces performance that is closely aligned with the SVM configuration, consistent with linear separability in the sparse feature space. Multinomial Naive Bayes exhibits lower recall in both folds. XGBoost displays marked variability, most notably in LOCO-2, where recall declines sharply. The embedding-based Jina v3 + MLP baseline remained competitive in LOCO-1 but showed reduced recall in LOCO-2, reinforcing the impact of strict case-level separation on cross-case transferability.

Table 5 reports the fold-level aggregation.

TABLE 5. Summary: Mean $\pm$ Std across LOCO folds.

Model	Acc Mean	Acc Std	F1 Mean	F1 Std
TF-IDF + LinearSVC	0.9929	0.0023	0.9828	0.0057
TF-IDF + LogReg	0.9905	0.000045	0.9769	0.000134
TF-IDF + MultinomialNB	0.9680	0.0108	0.9164	0.0303
TF-IDF + XGBoost	0.9138	0.0624	0.7006	0.2391
Jina v3 + MLP	0.9717	0.0188	0.9281	0.0486

The TF-IDF + LinearSVC configuration maintained the highest mean accuracy (0.9929) and F1 score (0.9828), with limited variation across folds. Logistic regression follows closely. In contrast, XGBoost exhibits substantial dispersion (F1 Std = 0.2391), indicating sensitivity to investigation-specific distributions. The embedding-based baseline shows moderate variability relative to the linear models.

Bootstrap resampling (1,000 iterations per fold) yielded 95% confidence intervals of [0.9822, 0.9989] and [0.9548, 0.9978] for accuracy and F1, respectively. These intervals remain narrow under LOCO separation, indicating that a single favorable split does not drive cross-case performance.

C. EFFICIENCY COMPARISON

End-to-end latency is defined as the interval from the in-memory raw text input to the final classification output. This includes text normalization, TF-IDF transformation, and evaluation of the SVM decision function for the TF-IDF + LinearSVC pipeline. For the embedding-based baseline, the measurement includes embedding generation followed by the forward pass of multilayer perceptrons. The inference-only latency isolates the classifier stage using precomputed feature vectors or embeddings. File input/output and document extraction are excluded.

The measurements were averaged over 100 iterations after a brief warm-up phase. Table 6 summarizes the results.

TABLE 6. Comparison of efficiency.

Model	Pipeline	Mean (ms)	p95 (ms)	Throughput	Size (MB)
TF-IDF + LinearSVC	End-to-End	0.317	0.375	3153.7	1.3400
TF-IDF + LinearSVC	Inference-only	0.001	0.001	1222715.3	1.3400
Jina v3 + MLP	End-to-End	3532.0	3885.2	0.283	2283.4
Jina v3 + MLP	Inference-only	0.007	0.008	145724.6	3.4100

The model size is reported as the on-disk footprint of each pipeline component. This includes all components required to process raw text input, including the embedding model, for end-to-end configurations. For inference-only configurations, only the classifier weights are reported because feature vectors are assumed to be precomputed. The TF-IDF + LinearSVC pipeline (vectorizer + linear weights) occupied 1.34 MB. In contrast, the embedding baseline required a local transformer checkpoint to generate Jina v3 embeddings, resulting in a multi-gigabyte footprint. By design, all latency measurements were CPU-only, reflecting the intended deployment setting of disconnected or resource-constrained investigative workstations.

These results highlight the practical implications of the architectural simplicity. The difference in latency and memory footprint directly affects throughput and operational scalability under resource-constrained deployment scenarios.

These differences are visually summarized in Fig. 2. The proposed TF-IDF + LinearSVC pipeline demonstrates orders-of-magnitude reductions in the end-to-end latency and the size of the deployed model (1.34 MB versus 2,283.4 MB) compared with the embedding-based baseline, while maintaining competitive inference-only performance. Such disparities directly affect deployment feasibility in resource-constrained investigative environments.

D. PROXY FALSE-POSITIVE EVALUATION

The trained SVM was applied to KorQuAD v1.0 [13], a publicly available Korean question-answering corpus composed of Wikipedia paragraphs, to estimate spurious activations under benign language conditions. All 9,606 unique context paragraphs exceeding 200 characters were included.

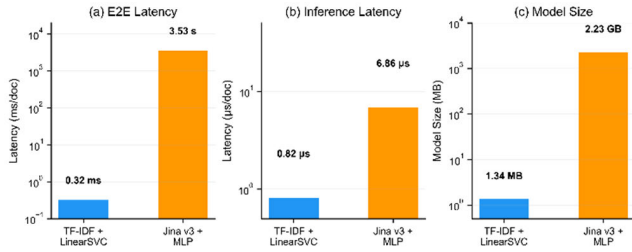


FIGURE 2. Efficiency comparison: (a) end-to-end latency, (b) inference-only latency, and (c) model size.

The length threshold was selected to approximate the document-length distribution of authentic COI texts and avoid trivial discrimination based on short fragments.

The classifier labeled 30 of 9,606 paragraphs as COI-positive, corresponding to a 0.3123% proxy false-positive rate. The Clopper–Pearson exact 95% confidence interval is [0.2108%, 0.4455%]. These values characterize activation behavior under out-of-domain text rather than in-domain investigative material.

The operational workload implications can be analytically expressed. Let N denote the number of seized documents and π the prevalence of COI documents within the corpus. For threshold τ , the expected number of flagged documents is as follows:

$$N [\pi \cdot \text{TPR}(\tau) + (1 - \pi) \cdot \text{FPR}(\tau)] \quad (7)$$

This expression translates the proxy false-positive rates into the expected review volume and complements Recall@K and WSS when selecting τ under operational constraints.

Robustness was further examined using a topically filtered hard-negative subset comprising 5,680 KorQuAD paragraphs containing keywords related to military, governmental, organizational, or directive themes. Within this subset, 16 false positives were observed (0.2817%, 95% CI: [0.1611%, 0.4570%]). The similarity between the full-set and hard-negative rates indicates that thematic overlap does not drive activation.

E. THRESHOLD SWEEP

Fig. 3 shows the relationship between recall on COI-positive documents and the proxy false-positive rate as the decision threshold τ varies. Increasing τ reduces the proxy false-positive rate and lowers recall.

The threshold is a tunable deployment parameter. Different operating points can be selected based on the acceptable review volume and investigative risk tolerance.

F. OPERATIONAL TRIAGE METRICS

As the framework operates as a rank-and-review system, the accuracy of aggregate classification does not fully capture operational behavior. To characterize the prioritization performance, Recall@K and WSS are reported.

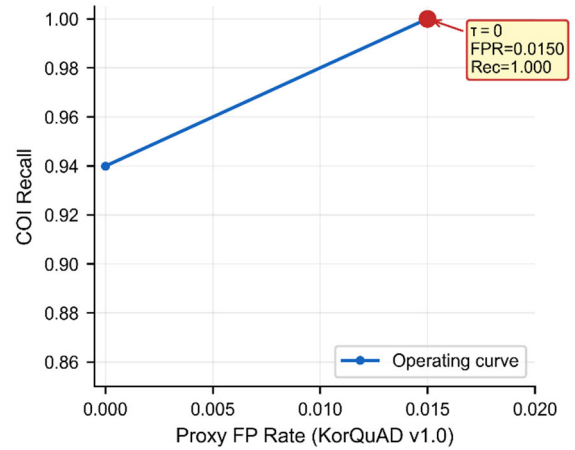


FIGURE 3. Recall versus proxy false-positive rate as a function of decision threshold τ . The default configuration ($\tau = 0$) is indicated.

TABLE 7. Operational triage metrics (mean across LOCO folds).

Target Recall	K (mean)	Corpus Reviewed	WSS
95%	64	20.0%	80.0%
100%	66	21.0%	79.0%

Table 7 presents the mean values across the LOCO folds.

At a target recall of 95%, a review of approximately 20% of the corpus is sufficient. Achieving complete recall requires the review of 21% of documents, corresponding to a WSS of 79%.

At $K = 50$, mean Recall@K reaches 78.1% with a Precision@K of 94.0%. These values indicate that COI-positive documents are concentrated near the top of the ranked list. The observed ranking behavior indicates that under the evaluated conditions, decision scores can be used to reduce review volume without sacrificing detection.

G. FEATURE ANALYSIS

We examine whether the discriminative features remain consistent across the investigations. If the model were primarily driven by case-specific expressions, cross-case transfer would deteriorate under LOCO separation.

Table 8 lists the top-20 TF-IDF features ranked by the absolute SVM weight. Among these, 12 of the 20 features appear in more than 20% of the documents in both investigations. Several features, particularly abbreviated or context-bound expressions, remain case-specific. However, a subset of directive constructions and coordination-related phrases is shared across cases.

To quantify alignment beyond frequency overlap, correlations between mean SHAP values for Cases A and B were computed. The Pearson correlation coefficient is $r = 0.364$ ($p < 0.001$), and Spearman's $\rho = 0.238$ ($p < 0.001$). These values do not imply the identity of feature importance across investigations; however, they are unlikely to arise from random variation alone.

Taken together, the overlap in high-weight lexical features and non-zero SHAP correlations are consistent with

TABLE 8. Top-20 TF-IDF Features with Cross-Case Distribution.

Rank	Feature	SVM Coef.	Case A (%)	Case B (%)	Cross-Case
1	tk	0.6421	0.0%	96.6%	No
2	관련한	0.6383	77.3%	70.8%	Yes
3	이와	0.6018	43.2%	87.6%	Yes
4	나갔으면 합니다	0.5524	2.3%	78.7%	No
5	나갔으면	0.5524	2.3%	79.8%	No
6	본사	0.5516	95.5%	93.3%	Yes
7	벌여	0.5192	15.9%	75.3%	No
8	비롯한	0.5065	47.7%	77.5%	Yes
9	경의	0.4722	6.8%	61.8%	No
10	q 에서는	0.4699	0.0%	59.6%	No
11	보냅니다	0.4434	65.9%	70.8%	Yes
12	맞게	0.4254	88.6%	67.4%	Yes
13	자료들	0.4198	31.8%	27.0%	Yes
14	보내주었으면	0.4192	2.3%	12.4%	No
15	보내주었으면 합니다	0.4173	2.3%	9.0%	No
16	관련하여	0.3823	61.4%	88.8%	Yes
17	것과	0.3820	25.0%	61.8%	Yes
18	것을	0.3816	86.4%	68.5%	Yes
19	주기	0.3758	84.1%	58.4%	Yes
20	이와 관련하여	0.3750	27.3%	77.5%	Yes

Note: For interpretive clarity, approximate English glosses of the top-ranked Korean features are provided: tk (abbreviation code), 관련한 (“related to”), 이와 (“with respect to this”), 나갔으면 합니다 (“we request that it proceed”), 본사 (“headquarters”), 벌여 (“to initiate”), 비롯한 (“including”), 경의 (“through”), 보냅니다 (“we send”), 맞게 (“appropriately”), 자료들 (“materials”), 보내주었으면 (“please send”), 관련하여 (“in relation to”), 것과 (“and the matter of”), 것을 (“the matter of”), 주기 (“cycle/period”), 이와 관련하여 (“in relation to this”).

the consistent lexical core hypothesis. Recurrent directive expressions appear across independent cases and are detectable using sparse linear representations.

H. ROBUSTNESS AND SENSITIVITY TO MODELS

Beyond the aggregate LOCO metrics, we examine the error distribution and parameter sensitivity. Across both folds, the TF-IDF + LinearSVC configuration produces five false negatives compared with eighteen for the embedding-based baseline. The higher miscount observed in LOCO-2 coincides with the smaller Case A training set (44 documents), limiting vocabulary exposure during training rather than implying decision boundary instability.

Sensitivity to vocabulary size is evaluated using feature-space ablation. As shown in Table 9, reducing the vocabulary from 35,996 to 5,000 terms does not alter accuracy or F1, while decreasing the model footprint and modestly reducing latency. The discriminative signal is concentrated in a restricted subset of features.

To evaluate the impact of Korean-specific preprocessing, Table 10 compares whitespace tokenization against Okt morphological analysis. Okt achieves perfect recall (F1 = 1.0,

TABLE 9. Vocabulary Ablation Size.

max_features	Features	Accuracy	F1	Size (MB)	Latency (ms)
5000	5000	0.9929	0.9828	0.2200	0.2763
10000	10000	0.9929	0.9828	0.4600	0.2857
20000	20000	0.9929	0.9828	0.9500	0.2964
Unlimited	35996	0.9929	0.9828	1.3400	0.3590

TABLE 10. Tokenizer ablation under LOCO evaluation.

Tokenizer	Mean F1	FN	Latency	Overhead
Whitespace	0.9828	5	0.37 ms	1×
Okt	1.0	0	53 ms	143×

Note: Whitespace tokenization is based on simple space-based splitting, whereas Okt denotes a Korean morphological analyzer. All experiments were conducted under the same LOCO protocol.

0 FN) by decomposing compound expressions into semantically meaningful units, but incurs 143× latency overhead (53 ms vs. 0.37 ms). For CPU-constrained deployment where sub-millisecond inference is prioritized, the whitespace tokenizer is retained as the default configuration.

Table 11 shows the explanation overhead. Weight-based top-*k* inspection adds approximately 0.26 ms per document. SHAP computation remains below 1 ms in the single-document setting due to the model’s linear structure. The added cost remains small relative to the full end-to-end latency.

TABLE 11. Explainability overhead.

Method	Mean (ms)	p95 (ms)	Notes
Fast Explanation (w ? x top-10)	0.263	0.332	Per-document, no model re-fit
SHAP	0.733	0.953	Per-document, includes explainer init
LinearExplainer (single) SHAP	0.189	N/A	Amortized over 633 docs
LinearExplainer (amortized)			

The potential reliance on synthetic artifacts is directly examined. None of the highest weight SVM features are exclusive to synthetic negatives. For example, the token “tk” appears in 64.7% of authentic COI documents and none of the synthetic texts. Removing all Latin-script tokens (113 in the full vocabulary; mean 59 per LOCO fold) and retraining under the same LOCO protocol left performance unchanged (F1 = 98.28%).

These observations are consistent with a model driven by a recurring Korean directive vocabulary rather than formatting artifacts introduced during the generation of synthetic data.

VI. DISCUSSION

The results indicate that when lexical regularities persist across investigations, cross-case forensic triage is feasible under sparse linear modeling. Under strict case-level separation, the performance remains consistent across independent datasets. This pattern suggests that discriminative cues are

not limited to investigation-specific phrasing alone. The proposed framework combines TF-IDF features, threshold-based ranking, and feature-level attribution. Together, these components define a system that remains computationally modest while preserving interpretability during review. The following discussion examines these observations from several perspectives: linguistic structure, model behavior, and deployment constraints.

A. EXAMINATION OF LEXICAL OVERLAP ACROSS CASES

The cross-case performance under LOCO separation remains consistently high for the linear classifier. This outcome is compatible with the presence of a recurring lexical structure in the COI documents. Accuracy above 99% under investigation-level separation implies that classification does not solely depend on shared phrasing within a single case.

This should not imply that the task itself lacks complexity. Directive texts in this domain are bureaucratic and structurally constrained. Recurrent rhetorical templates and standardized expressions limit lexical dispersion. Nevertheless, variation persists. Investigation-specific operational terminology appears alongside a more consistent formulaic coordination language layer.

As reported in Table 8, 12 of the top-20 SVM-weighted features occur in more than 20% of documents in both investigations. The pattern reflects partial regularity: structural persistence combined with contextual differentiation.

B. ERROR ANALYSIS

The LinearSVC triage engine produced 5 false negatives across both LOCO folds: 1 in LOCO-1 (FNR = 2.27%, 1/44) and 4 in LOCO-2 (FNR = 4.49%, 4/89). Table 12 summarizes the false-negative profile per fold. Qualitative examination of these missed documents reveals two recurring patterns. First, three of the five false negatives exhibited attenuated directive rhetoric. Although they opened with institutional greetings such as “뜨거운 동지적 인사를 보냅니다” (warm comradely greetings), the body text shifted toward administrative or technical content—requests for livestock breeding references, organizational status inquiries, or scheduling coordination—that diluted the lexical density of directive-specific vocabulary. The TF-IDF representation assigned lower aggregate scores because high-IDF directive markers appeared only in the salutation. Second, two false negatives were notably shorter than the corpus median and contained fewer n-gram features overall, reducing the cumulative margin for positive classification. Decision scores for all five documents clustered within ± 0.13 of the zero threshold (range: -0.126 to -0.006), indicating borderline cases rather than confident misclassifications. From an operational standpoint, the observed overall FNR of 3.76% (5/133) implies that approximately one in twenty-seven COI documents may bypass triage. In high-stakes investigations, analysts may lower the decision threshold (e.g., $\tau = -0.3$) to recover additional positives at the cost of increased false-positive volume.

TABLE 12. False-negative profile per LOCO fold.

Fold	Test Positives	FN	FNR (%)	Score Range
LOCO-1	44	1	2.27	-0.006
LOCO-2	89	4	4.49	-0.126, -0.014
Overall (pooled)	133	5	3.76	—

C. EXPLAINABILITY SCOPE OF SHAP

SHAP explanations in this framework serve a strictly defined operational function. For each document classified as COI-positive, SHAP provides an additive decomposition of the model score into per-feature contributions, enabling the analyst to identify which lexical patterns drove the classification. This output is designed as analyst-facing rationale: it supports review prioritization and helps the analyst assess whether the flagged document warrants further examination.

SHAP values do not constitute a standalone forensic opinion, an evidentiary analysis, or a substitute for expert review of the original document. They reflect model behavior under the trained TF-IDF representation; they do not perform semantic interpretation, assess context, or evaluate document authenticity. In any legal or investigative proceeding, the triage score and associated SHAP rationale must be treated as one input within a broader human-led review process. Chain-of-custody documentation, original document analysis, and expert testimony remain primary.

This framework is a screening aid within a tiered forensic workflow, not an autonomous adjudicative instrument.

D. LINEAR MODELS VERSUS DEEP LEARNING BASELINES

The embedding-based Jina v3 + MLP configuration exhibits lower cross-case recall in LOCO-2 (82.02%) compared with linear models (mean accuracy, 97.17%). One interpretation concerns representation smoothing. The transformer encoders map lexically distinct tokens into semantically proximate vectors. Such smoothing may attenuate the influence of rare but discriminative expressions that contribute to cross-case separation in domains characterized by rigid and repetitive directive markers.

The XGBoost baseline shows marked degradation in LOCO-2 (accuracy 85.14%, F1 46.15%). This asymmetry aligns with the known behavior of tree-based ensembles under limited positive samples and high-dimensional sparse inputs. With 44 positive training instances and a feature space exceeding 10,000 dimensions, boosted trees may form partitions that do not generalize but are anchored to case-specific splits.

An L1-regularized LinearSVC was evaluated to determine whether extreme sparsification preserves transferability. The full-corpus L1 model retained 19 nonzero features (99.9% sparsity), and the LOCO fold models retained 20 and 10 features, respectively. LOCO-1 accuracy remained at 97.6%, whereas LOCO-2 F1 accuracy decreased to 51.2%. Aggressive elimination of feature appears to isolate investigation-specific indicators rather than recurring lexical structure.

The L2-regularized linear configuration maintains broader lexical coverage. This broader representation under LOCO separation supports more consistent cross-case transfer.

E. OPERATIONAL CONSIDERATIONS

The TF-IDF + LinearSVC configuration was constructed under deployment constraints. Its computational profile allows full-corpus screening as a routine preliminary stage rather than a selective subset operation. SHAP-based explanations provide traceable feature attribution for flagged documents and support documentation requirements in evidentiary contexts. The adjustable decision threshold further enables the selection of operating points that are aligned with investigative priorities.

Documents located near the decision boundary may warrant secondary review. In such instances, a more complex model could be applied selectively within a tiered workflow [5], [6], [18]. This arrangement preserves the efficiency of the initial screening stage while reserving computationally intensive analysis for cases with the highest uncertainty.

F. LIMITATIONS

The proposed framework relies on surface-level lexical representations and therefore inherits the constraints of feature-based text triage.

One limitation concerns the use of synthetic negatives. The primary claim of cross-case lexical regularity is derived solely from authentic COI documents; however, the synthetic class may not reflect the full distribution of benign seized material. Inspection of high-weight features revealed no tokens exclusive to synthetic texts. Evaluation of 9,606 independent KorQuAD paragraphs produced a low proxy false-positive rate with narrow confidence bounds, and the results were similar for a topically filtered subset. These checks reduce—but do not eliminate—the possibility of distributional mismatch. Broader evaluation using authentic benign corpora would strengthen external validity.

The dataset contains 133 documents from two investigations conducted between 2018 and 2022. LOCO validation confirms transfer across these cases; however, institutional language may evolve over longer time horizons. Feature distributions could shift accordingly. Periodic retraining is required under such conditions.

The framework was developed for directive texts within national security investigations. Extension to other document genres has not been established. The tokenizer operates at the surface form level without morphological segmentation. This is sufficient for the structured corpus examined here, but corpora with greater lexical variability may require finer grained analysis.

Finally, this study does not address adversarial obfuscation strategies. Deliberate paraphrasing, systematic synonym substitution, steganography, or other encryption methods may reduce detection reliability. The system is intended as a screening aid within a tiered forensic workflow and not as a standalone adjudication instrument.

G. EXTERNAL VALIDITY

The reported results are bounded by four explicit validity constraints:

(1) Language: The framework is validated exclusively on Korean-language documents. Lexical transfer to other languages has not been evaluated and cannot be assumed. (2) Document type: Validation is limited to counterintelligence-type directive documents (COI) characterized by institutional command language. Extension to other crime types—financial fraud, narcotics trafficking, cybercrime—has not been assessed.

(3) Jurisdiction: Both source investigations fall under the Korean National Security Act. Applicability to other legal frameworks or investigative contexts is not established.

(4) Temporal scope: Training and evaluation documents span 2018–2022. Lexical drift over longer time horizons may degrade performance.

These boundaries define the operational envelope within which the proof-of-feasibility holds. Deployment outside this envelope requires dedicated validation with representative in-domain data.

H. OBFUSCATION ROADMAP

The current framework assumes that COI documents retain their characteristic lexical surface forms. Adversarial actors aware of automated screening may attempt to evade detection through deliberate obfuscation. Four threat categories warrant future investigation:

(1) Synonym and paraphrase substitution: replacing directive-specific terms (e.g., “활동방향” → “업무지침”) to reduce TF-IDF feature overlap with training exemplars. (2) Homoglyph replacement: substituting visually similar Unicode characters (e.g., Cyrillic “à” for Latin “a”) to break exact n-gram matching while preserving human readability. (3) Encoding-based concealment: embedding directive content within Base64, hexadecimal, or other reversible encodings that render surface text opaque to bag-of-words extraction. (4) Steganographic embedding: hiding directive payloads within carrier documents (e.g., whitespace encoding, zero-width characters) such that the visible text appears benign.

Threats (1) and (2) may be partially addressed within the existing lightweight architecture through character-level normalization and synonym-aware feature expansion. Threats (3) and (4) require preprocessing modules capable of detecting and decoding concealed content—an extension compatible with CPU-only deployment but demanding additional engineering. Robust evaluation against these obfuscation strategies constitutes a priority direction for subsequent work.

VII. CONCLUSION

A lightweight forensic triage framework was developed to prioritize clandestine operational instruction documents under CPU-only deployment constraints. The pipeline combines TF-IDF sparse representation, a threshold-tunable

LinearSVC classifier, SHAP-based analyst rationale, and structured reporting within a single auditable architecture.

Under strict leave-one-case-out separation across two independent, court-adjudicated investigations, the linear configuration maintained high recall with zero false positives and a pooled false-negative rate below 4%. Comparison against fastText and KoELECTRA-small indicated that the proposed configuration offers a favorable balance of classification performance, latency, model footprint, and feature-level audibility among the evaluated CPU-deployable alternatives. A preprocessing ablation showed that morphological analysis improves F1 but incurs latency overhead that conflicts with the sub-millisecond design target, supporting the whitespace tokenization choice.

These results suggest that cross-case triage is feasible under sparse linear modeling when the target documents share recurring institutional directive language. The present evidence is confined to two Korean-language counterintelligence investigations and does not establish validity across other crime types, languages, or jurisdictions. Future work will address obfuscation-aware extension, evaluation across additional adjudicated cases, and multilingual adaptation within the same lightweight deployment family.

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